

AI Application Lecture 1

Real-World Problems and the Formulation of Machine Learning

SUZUKI, Atsushi^a

^aI thank my collaborator, Jing WANG, for cooperating in creating the teaching materials.

Outline

1. Introduction
2. Preliminaries: Mathematical Notations
3. What is AI? - Task Solving Machine
4. Can We Formulate Various Tasks Uniformly? - As Input-Output Relationships
5. Can Computers Input and Output Real Things? - Numerical Representation
6. What is a computer-friendly formulation of constructing AI? - As a function determination
7. Summary and Future Outlook

Introduction: Purpose

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The purpose of this course is to learn the fundamentals of generative AI in the application context from scratch. For example, neural networks in this course do NOT refer to multilayer perceptrons, but those with a flexible graph structure, including Transformer or U-net.

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This course will start with **fundamental** topics such as the definitions of machine learning and neural networks.

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Reason: The field of AI and machine learning is developing very rapidly, so specific technologies and models that are widely known today **may well be obsolete by the time you graduate.**

Note: “fundamental” does not mean “theory.” Rather, this course tries to study general and universal concepts applicable to diverse applications.

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1. First, we will treat AI as a functional **black box** and start by learning "how to use it."
2. Then, as necessary, we will **white-box** the contents of the black box and uncover its mechanisms.

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- **Acquiring practical knowledge:** You will at least acquire the practical knowledge of "how to use" AI, even if you give up on the way.
- **Gaining universal knowledge:** The "framework for using AI" is relatively resistant to change.
- **Addressing realistic constraints:** Details of large-scale models are often trade secrets, and there is no "panacea learning method for all problems."

Learning Outcomes

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- Formulate tasks that humanity seeks to solve as a relationship between **input** and **output**.
- Mutually convert various real-world **entities** that humanity deals with and their corresponding **numerical sequences**.
- Formulate constructing an **Artificial Intelligence** system as a **function determination problem**.

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Preliminaries: Mathematical Notations

We will organize the basic mathematical notations used in this lecture.

- **Definition:**
 - $(\text{LHS}) := (\text{RHS})$: Indicates that the left-hand side is defined by the right-hand side.
 - For example, $a := b$ indicates that a is defined as b .

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 - For example, $a := b$ indicates that a is defined as b .
- **Set:**
 - Sets are often denoted by uppercase calligraphic letters. E.g., $\mathcal{A}$.
 - $x \in \mathcal{A}$: Indicates that the element x belongs to the set $\mathcal{A}$.
 - $\{\}$: The empty set.

Preliminaries: Mathematical Notations

- **Set Notation Examples:**

- $\{a, b, c\}$: The set consisting of elements a, b, c (Roster notation).
- $\{x \in \mathcal{A} | P(x)\}$: The set of elements in $\mathcal{A}$ for which the proposition $P(x)$ is true (set-builder notation).

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- **Common Sets of Numbers:**

- $\mathbb{R}$: The set of all real numbers.
- $\mathbb{R}_{>0}$: The set of all positive real numbers.
- $\mathbb{R}_{\geq 0}$: The set of all non-negative real numbers.
- $\mathbb{Z}$: The set of all integers.
- $\mathbb{Z}_{>0}$: The set of all positive integers.
- $\mathbb{Z}_{\geq 0}$: The set of all non-negative integers.
- $[1, k]_{\mathbb{Z}} := \{1, 2, \dots, k\}$: The set of integers from 1 to k .

Preliminaries: Mathematical Notations

- **Function:**

- $f : \mathcal{X} \rightarrow \mathcal{Y}$: f is a **map** from a **domain** $\mathcal{X}$ to a **codomain** $\mathcal{Y}$.
- $y = f(x)$: The output of f for an input $x \in \mathcal{X}$ is $y \in \mathcal{Y}$.

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- **Vector:**

- In this course, a vector is a column of numbers.
- Denoted by bold italic lowercase letters (e.g., $\mathbf{v}$).
- $\mathbf{v} \in \mathbb{R}^n$: an n -dimensional real vector.
- The i -th element is denoted as v_i .

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Preliminaries: Mathematical Notations

- **Matrix:** Denoted by bold italic uppercase letters (e.g., $\mathbf{A}$).
 - $\mathbf{A} \in \mathbb{R}^{m,n}$: an $m \times n$ real matrix.
 - The element in the i -th row and j -th column is denoted as $a_{i,j}$.

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$$

Preliminaries: Mathematical Notations

- **Matrix Transpose:** The transpose of A is denoted as $A^\top$.
 - If $A \in \mathbb{R}^{m,n}$, then $A^\top \in \mathbb{R}^{n,m}$.

$$A^\top = \begin{bmatrix} a_{1,1} & a_{2,1} & \cdots & a_{m,1} \\ a_{1,2} & a_{2,2} & \cdots & a_{m,2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1,n} & a_{2,n} & \cdots & a_{m,n} \end{bmatrix}$$

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- **Vector Transpose:** A vector is a matrix with 1 column.
 - The transpose of a column vector is a row vector.

$$\mathbf{v}^\top = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \in \mathbb{R}^{1,n}$$

Preliminaries: Mathematical Notations

- **Tensor:**
 - In this lecture, we treat tensors as multidimensional arrays.
 - A vector is a 1st-order tensor; a matrix is a 2nd-order tensor.
 - Tensors of 3rd order or higher are denoted by an underlined, bold, italic, uppercase letter, e.g., $\underline{\mathbf{A}}$.
 - Note: For those familiar with abstract tensors, we assume a fixed standard basis, identifying a tensor with its component representation.

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 - Note: For those familiar with abstract tensors, we assume a fixed standard basis, identifying a tensor with its component representation.
- **Note on Implementation:**
 - Theoretically, we use real numbers ($\mathbb{R}$).
 - In practice, computers use finite subsets (e.g., floating-point numbers).
 - These subsets are approximations. Their discreteness and finiteness can cause different properties from $\mathbb{R}$.

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7. Summary and Future Outlook

Question: What is AI? Or,

Question: What is AI? Or, how should we define AI?

Defining AI is not Easy

This course is about "AI applications," but what is "Artificial Intelligence" in the first place? Defining this term is extremely difficult.

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- **Intelligence:** No unified definition exists, even among experts.

Defining AI is not Easy

This course is about "AI applications," but what is "Artificial Intelligence" in the first place? Defining this term is extremely difficult.

- **Artificial:** Realized using a **computer**.
- **Intelligence:** No unified definition exists, even among experts.

From an engineering perspective, what is important is whether that technology can contribute to human society.

What is AI?

This course will not delve into any philosophical discussions, but will define AI in a very practical way as follows.

Definition (Definition of AI in This Lecture)

Artificial Intelligence (AI) is a procedure implemented on a computer that solves some task that humanity wants to solve.

What is AI? - Our answer

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Answer:

What is AI? - Our answer

Question: What is AI? Or, how should we define AI?

Answer: AI is defined in this lecture as a procedure implemented on a computer that solves some task that humanity wants to solve.

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Can we formulate those various tasks in a unified manner?

Task Formulation: Problems as Input-Output Relationships

Many of the tasks that humans want computers to solve can be formulated as a problem of "returning an appropriate output given a certain input."

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This is the idea of viewing a task as a kind of transformation rule, that is, a correspondence between input and output. Let's look at some examples.

Task Formulation: Examples

- **Chat AI:**

- **Input:** A question or command sentence. (e.g., "What's the weather like today?")
- **Output:** The AI's response text. (e.g., "The weather in Tokyo is sunny.")

Task Formulation: Examples

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- **Text-to-Image (Image Generation AI):**

- **Input:** A descriptive sentence (**prompt**). (e.g., "A cat flying in space, oil painting style")
- **Output:** Generated image data.

Task Formulation: Examples

- **Medical Diagnosis from Images:**
 - **Input:** Medical image data (X-rays, CT scans, etc.).
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As can be seen from these examples, even tasks that seem completely different share a common structure of "input → output."

Task Formulation - Answer

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Task Formulation - Answer

Question: We want AI to solve various tasks.

Can we formulate those various tasks in a unified manner?

Answer: Most tasks are a process of **returning an appropriate output given a certain input.**

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How can a computer/AI

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How can a computer/AI deal with real-world input and output?

Numerical Representation of Real-World Entities

Fundamentally, what computers can handle are **bit strings** of 0s and 1s, or the **byte strings** and numbers they form.

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Fundamentally, what computers can handle are **bit strings** of 0s and 1s, or the **byte strings** and numbers they form.

Therefore, to process the various inputs and outputs we saw in the previous section on a computer, they must first be converted into numerical data. This conversion rule is defined by humans according to the task.

Natural Language

A sentence is a sequence of characters. The rule that maps each character to a specific byte sequence is called a **character encoding**.

- **ASCII (American Standard Code for Information Interchange):**
 - Maps alphabets, numbers, symbols, etc., to 7-bit (usually 1-byte) integers.
 - For example, 'A' is 65, and 'B' is 66.

Natural Language: ASCII Table

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
32	20	(space)	56	38	8	80	50	P	104	68	h
33	21	!	57	39	9	81	51	Q	105	69	i
34	22	"	58	3A	:	82	52	R	106	6A	j
35	23	#	59	3B	;	83	53	S	107	6B	k
36	24	\$	60	3C	<	84	54	T	108	6C	l
37	25	%	61	3D	=	85	55	U	109	6D	m
38	26	&	62	3E	>	86	56	V	110	6E	n
39	27	'	63	3F	?	87	57	W	111	6F	o
40	28	(	64	40	@	88	58	X	112	70	p
41	29	)	65	41	A	89	59	Y	113	71	q
42	2A	*	66	42	B	90	5A	Z	114	72	r
43	2B	+	67	43	C	91	5B	[	115	73	s
44	2C	,	68	44	D	92	5C	\	116	74	t
45	2D	-	69	45	E	93	5D	]	117	75	u
46	2E	.	70	46	F	94	5E	^	118	76	v
47	2F	/	71	47	G	95	5F	_	119	77	w
48	30	0	72	48	H	96	60	`	120	78	x
49	31	1	73	49	I	97	61	a	121	79	y
50	32	2	74	4A	J	98	62	b	122	7A	z
51	33	3	75	4B	K	99	63	c	123	7B	{
52	34	4	76	4C	L	100	64	d	124	7C	
53	35	5	77	4D	M	101	65	e	125	7D	}
54	36	6	78	4E	N	102	66	f	126	7E	~
55	37	7	79	4F	O	103	67	g			

Natural Language: UTF-8

- **UTF-8 (Unicode Transformation Format-8):**
 - Includes characters from all over the world.
 - Uses a variable-length byte sequence (1 to 4 bytes).
 - e.g., 'A' is 1 byte, while '𐀀' and '𐀀𐀀'¹ are 3 and 4 bytes.

¹ U+1F602 [<https://www.utf8-chartable.de/unicode-utf8-table.pl?start=127744&number=1024>] (<https://www.utf8-chartable.de/unicode-utf8-table.pl?start=127744&number=1024>)

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Example (UTF-8 representations of letters and characters)

- '𐄀' (U+65E5) → UTF-8 (hex): E6 97 A5 (3 bytes)
- Greek 'Ω' (U+03A9) → UTF-8 (hex): CE A9 (2 bytes)
- Emoji '𐄀𐄀' (U+1F602) → UTF-8 (hex): F0 9F 98 82 (4 bytes)

With this rule, any text can be converted into a unique byte sequence.

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Natural Language: Remark on Emoji

Remark (On the Technical Significance of Emoji)

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Emoji, originating from Japan, are now used worldwide. Their popularization has motivated worldwide support for multi-byte characters.

For instance, people in the Chinese character culture sphere (China, Japan, Korea, Vietnam) have benefited from this, as it improved support for characters like Chinese characters. Before emoji became popular, supporting these character sets was not a high priority for the predominantly English-speaking tech world.

Natural Language: Remark on Emoji

Remark (On the Technical Significance of Emoji)

The word Emoji comes from the Japanese word "絵文字" (e-moji), meaning "picture character," not from "emotion." The practice of assigning character codes to pictures dates back to the 1950s in Japan.

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However, the contribution of the Gmail team in assigning Unicode points was also significant for the global spread of emoji.

Images

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- **Bitmap representation:**

- An image of height H and width W can be represented as a 3rd-order integer tensor $\underline{I} \in \mathbb{Z}^{3 \times H \times W}$.
- Each of the three $H \times W$ matrices corresponds to the R, G, and B channels, with integer values from 0 to 255.

Images: Example

Consider a small 2×2 pixel image:

- Top-left: Red (R:255, G:0, B:0)
- Top-right: Green (R:0, G:255, B:0)
- Bottom-left: Blue (R:0, G:0, B:255)
- Bottom-right: White (R:255, G:255, B:255)

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This image is represented as a $3 \times 2 \times 2$ tensor $\underline{\mathbf{I}}$, where the channels $\underline{\mathbf{I}}_R, \underline{\mathbf{I}}_G, \underline{\mathbf{I}}_B$ are:

$$\underline{\mathbf{I}}_R = \begin{bmatrix} 255 & 0 \\ 0 & 255 \end{bmatrix}, \quad \underline{\mathbf{I}}_G = \begin{bmatrix} 0 & 255 \\ 0 & 255 \end{bmatrix}, \quad \underline{\mathbf{I}}_B = \begin{bmatrix} 0 & 0 \\ 255 & 255 \end{bmatrix}$$

Video & Audio

- **Video:** A sequence of time-varying images.
 - Add a time dimension to the image tensor.
 - A video of T frames is a 4th-order tensor $\underline{\mathbf{V}} \in \mathbb{Z}^{3 \times T \times H \times W}$.

Video & Audio

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 - A video of T frames is a 4th-order tensor $\underline{\mathbf{V}} \in \mathbb{Z}^{3 \times T \times H \times W}$.
- **Audio:** A time-varying wave.
 - **PCM (Pulse Code Modulation):** The analog signal is converted to a sequence of integers through:
 - **Sampling:** Measuring amplitude at regular time intervals.
 - **Quantization:** Approximating each measurement with an integer value.

Audio: PCM

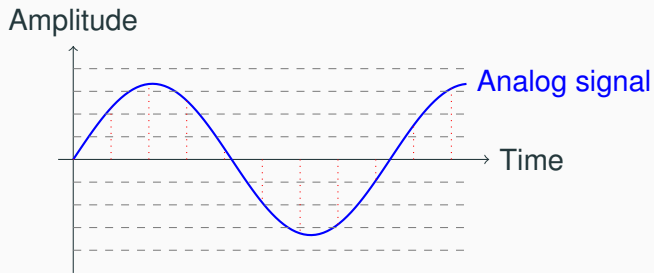


Figure 1: Schematic of sampling and quantization of an audio signal

Audio: PCM

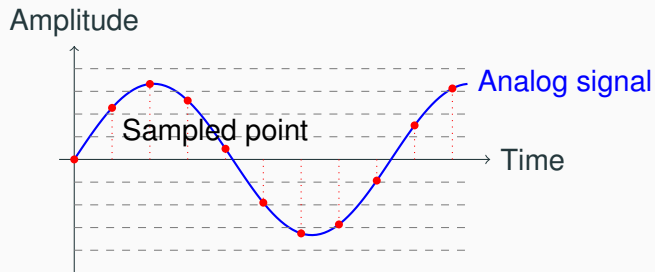


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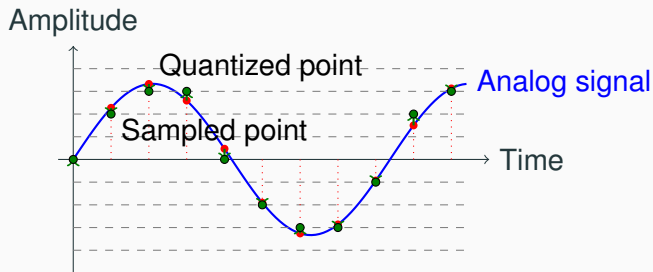


Figure 1: Schematic of sampling and quantization of an audio signal

Numerical Representation: Summary

In the case of stereo audio, this process is performed for the left and right channels respectively, resulting in two integer sequences $\mathbf{l}, \mathbf{r} \in \mathbb{Z}^D$.

Numerical Representation: Summary

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In this way, various real-world entities can be converted into numerical data (vectors or tensors) based on clear rules.

Numerical Representation - Answer

Question: How can a computer/AI deal with real-world input and output?

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Question: How can a computer/AI deal with real-world input and output?

Answer: We convert the real-world input and output to/from numerical data, using suitable methods depending on the data type.

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1. Introduction
2. Preliminaries: Mathematical Notations
3. What is AI? - Task Solving Machine
4. Can We Formulate Various Tasks Uniformly? - As Input-Output Relationships
- 5. Can Computers Input and Output Real Things? - Numerical Representation**
6. What is a computer-friendly formulation of constructing AI? - As a function determination
7. Summary and Future Outlook

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Now, let's summarize the discussion so far.

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1. The task we want to solve can be seen as a correspondence of "input entity $\rightarrow$ output entity."
2. Input and output entities can be converted into "input numerical data $\rightarrow$ output numerical data" according to appropriate rules.

Through these two steps, the problems we want AI to solve ultimately come down to the problem of "calculating the appropriate output numerical data given some input numerical data."

Redefining AI: The Function Determination Problem

This is none other than the **problem of finding a function** from the set of all input numerical data (**input space**) to the set of all output numerical data (**output space**).

Definition (Engineering Goal of AI)

To solve a task is to construct a function f on a computer that mimics an unknown ideal function $f^* : \mathcal{X} \rightarrow \mathcal{Y}$ that represents the input-output relationship of that task.

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- **Chat AI:**
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- **Text-to-Image:**

- Input space $\mathcal{X}$: Set of byte strings (prompts).
- Output space $\mathcal{Y} = \mathbb{Z}^{3 \times H \times W}$: Set of image tensors.

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Thus, we can see that any task can be uniformly described as a problem of determining a function f from an input space $\mathcal{X}$ to an output space $\mathcal{Y}$.

Formulation of Constructing AI - Answer

Question: What is a computer-friendly formulation of the “finding an appropriate numerical input-output relation” problem? Here, the formulation should be mathematical so that the computer can understand.

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Formulation of Constructing AI - Answer

Question: What is a computer-friendly formulation of the “finding an appropriate numerical input-output relation” problem? Here, the formulation should be mathematical so that the computer can understand.

Answer: Constructing an AI is a function determination problem.

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Today's Summary

In this lecture, we have organized the diverse tasks that AI tackles from a mathematical perspective.

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In this lecture, we have organized the diverse tasks that AI tackles from a mathematical perspective.

- **Formulating tasks as input-output:** Many tasks can be formulated as determining an output for a given input.
- **Numerical representation of entities:** Real-world entities are converted into computer-manageable numerical data (vectors and tensors).
- **AI and the function determination problem:** Solving a task with AI is equivalent to determining an appropriate function $f : \mathcal{X} \rightarrow \mathcal{Y}$.

Next Time

We have seen that solving a task can be equated with "determining a function."
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In the upcoming lectures, we will learn about neural networks, which are the concrete components of this function f , and uncover how they learn complex input-output relationships.